

- (A) Let $X : \Omega_1 \rightarrow \Omega_2$ be a $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ measurable transformation. Consider the collection (of subsets of Ω_1) defined by $\sigma\langle X \rangle = \{X^{-1}(B), B \in \mathcal{F}_2\}$. Then show that
- (a) $\sigma\langle X \rangle$ is a σ -algebra,
 - (b) $\sigma\langle X \rangle$ is the *smallest* σ -algebra that makes X measurable,
(i.e. if X is $\langle \mathcal{G}, \mathcal{F}_2 \rangle$ -measurable, then $\sigma\langle X \rangle \subset \mathcal{G}$).
- (B) Give an example of a function f that is **not** $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.